KNECK OFFICIAL

Geography — Paper 1

312/1 • ORIGINAL REVISION SET 01 • 2 hours 45 minutes • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all Section A questions. In Section B answer question 6 and ANY TWO of questions 7–10. Use supplied schematic map data; no external map is required.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SUPPLIED MATERIAL / RESOURCE

SCHEMATIC MAP DATA (original coordinate exercise, not an official Survey of Kenya map). Grid spacing is 1 km; scale 1:50,000. Eastings and northings are kilometres. To form a six-figure reference, use two-digit grid square numbers followed by tenths.
Feature coordinates: School S (21.4,34.6); Market M (24.4,38.6); Bridge B (24.0,36.0); Spring P (26.2,39.1). North is increasing northing; east is increasing easting. Road S–M is straight. River flows from P through B towards (22.0,33.0).
Cross-section on line west→east: horizontal distance km 0,1,2,3,4; altitude m 900,1100,1500,1300,1000.
A farm at altitude 1000 m has mean temperature 24°C. Use lapse rate 0.6°C per 100 m.
Rainfall data (mm), January→December: 40,50,120,180,140,60,40,50,70,130,110,60.

SECTION A

All questions; 25 marks.

1. Distinguish weather from climate and give three elements measured at a weather station. [5 marks]

2. Define plate tectonics and explain one way a fold mountain forms. [5 marks]

3. Name two intrusive igneous features and explain formation of one. [5 marks]

4. State two causes of river deposition and three resulting landforms. [5 marks]

5. Give five measures to reduce soil erosion on a cultivated slope. [5 marks]

SECTION B

Question 6 compulsory; then choose any TWO of questions 7–10. Each 25 marks.

6. Using supplied map data: (a) Give six-figure references of School and Market. (4) (b) Find straight distance S–M in km and corresponding map distance in cm. (4) (c) Calculate bearing of M from S to nearest degree. (4) (d) State general river flow direction and evidence. (3) (e) Draw the cross-section to suitable scales. (6) (f) Calculate mean gradient from distance 0 to 2 km as a ratio. (4) [25 marks]

7. (a) Explain formation of relief rainfall with a labelled diagram. (10) (b) Draw a monthly rainfall bar graph using supplied data. (6) (c) Calculate annual rainfall and identify wettest month. (4) (d) Estimate temperature at 1500m given the farm data and explain two limitations. (5) [25 marks]

8. (a) Describe formation of a waterfall and gorge. (8) (b) Explain three processes of river erosion. (6) (c) Describe three features of a meander. (6) (d) Explain five benefits or risks of floodplains to people. (5) [25 marks]

9. (a) Explain physical weathering by freeze-thaw and exfoliation. (8) (b) Explain chemical weathering by carbonation and hydrolysis. (6) (c) Give three factors influencing mass movement. (6) (d) State five slope-management measures. (5) [25 marks]

10. (a) Describe formation of a corrie and an arête. (10) (b) Explain two differences between glacial and river valleys. (4) (c) Describe three karst surface features. (6) (d) State five reasons for protecting mountain environments. (5) [25 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (5 marks) Weather short-term atmospheric conditions (1), climate long-term average/pattern normally decades (1); temperature,rainfall,pressure,humidity,wind,sunshine any three (3).

2. (5 marks) Lithospheric plates move over underlying mantle (2); convergence/compression of sediments, folding and uplift at collision margins (3).

3. (5 marks) Dyke/sill/batholith/laccolith, any two (2). Magma intrusion into cracks/layers, cooling/crystallisation beneath surface, matching geometry of chosen feature (3).

4. (5 marks) Reduced gradient/velocity, reduced discharge, increased load any two (2); floodplain,levee,delta,alluvial fan,point bar any three (3).

5. (5 marks) Terracing,contour farming,cover crops,mulching,agroforestry,grassed waterways,controlled grazing; any five (5).

SECTION B

6. (25 marks) a 214346;244386 (4). b ΔE=3,ΔN=4; distance5km, map10cm (4). c tanθ=3/4; bearing037° (4). d Generally southwest, decreasing easting/northing (3). e Labelled axes/scales 2; five points 3; joined profile1 (6). f Rise600m/run2000m; gradient600:2000=1:3.33 (4).

7. (25 marks) a Moist air forced upslope, cools by expansion, condenses at dew point, cloud/rain windward, drier descending leeward/rain shadow; explanation7 diagram3 (10). b axes/scale2,bars3,labels1 (6). c Total1050mm; April180mm (4). d Rise500m→3°C fall;21°C (3). Lapse varies with humidity/weather/inversion/exposure; two limitations (2).

8. (25 marks) a Resistant over softer rock, differential erosion, plunge pool/undercut, collapse, retreat/gorge with coherent sequence (8). b Hydraulic action,abrasion,solution,attrition, any three process explanations2 each (6). c Outer cut bank,inner slip-off slope,lateral migration/oxbow development, three2 each (6). d Fertile soils,water,transport/settlement, flooding/property loss,disease/erosion; five distinct (5).

9. (25 marks) a Water cracks/freezes/expands/repeats; temperature changes/pressure release and outer layers detach with relevant qualification (8). b Weak carbonic acid dissolves limestone; water reacts with minerals forming new minerals/clays (6). c Gradient,water loading,rock structure/vegetation/human disturbance three explained (6). d Drainage,vegetation,terracing,retaining structures,avoid undercutting/control construction; five (5).

10. (25 marks) a Snow accumulation/compaction, rotational ice/abrasion/plucking, overdeepened hollow/lip; adjacent corries erode back leaving sharp ridge (10). b U versus V cross-section; truncated versus interlocking spurs, paired (4). c Limestone pavement/clints-grikes,dolines/swallow holes,dry valleys etc, three explained (6). d Water catchments,biodiversity,soil stability,tourism,cultural value,carbon storage; five (5).

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf